Project 2: Milestone 3

Final Paper

Creating Third Shot Strategy in Pickleball

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Business Problem

The game of pickleball has evolved over the last several years. What was once a game reserved for the elderly or elementary gym classes has become the fastest growing sport with huge investment from both professional leagues and amateur players alike.

As the stakes become higher, prize money gets larger, and the amateur game becomes more athletic and skilled investment in analytic strategy is sure to increase.

Pickleball has quickly become a multibillion-dollar industry and shot selection strategy has the potential to disrupt the industry and provide a ton of value to both professionals and amateurs who have huge financial incentive or investment in their performance outcomes.

Background/ History

The status quo in pickleball has been that the strategy for the third shot of a rally is to perform a "third shot drop" or a shot that lands softly close to the net from the serving team to help neutralize the advantage the returning team has in pickleball of controlling the net from the beginning of the rally.

As the game has evolved to become more athletic by attracting players with more athletic backgrounds and the introduction of superior equipment that makes shots that were previously impossible possible it is time to reevaluate the strategy around the third shot. We have already seen players at the professional and high-level amateur range adopt more of a drive first strategy in some cases. This project will train a model to help players develop a shot selection strategy for the third shot that gives them the best opportunity to win the point given the conditions of the game. For example, the location and velocity of the return, the demographics of their team and the opponents as well as what point of the game the rally takes place. A model like this has the potential to maximize wins and improve player performance. With pickleball being a multibillion dollar industry there is and will be serious demand for analytics that can help player perform better in tournaments and matches of every level.

Data Explanation

Data for this project was collected from Pklmart. Pklmart is a pickleball analytic platform that breaks down matches for both amateurs and professionals. Pklmarts true power comes in their ability to track shot level data for all the matches loaded into the database. The database contains 8 tables and hundreds of columns of features. For this project Data will be pulled from 3 tables, the rally, shot and player tables.

The rally table will be used to collect the rally winner as well as the players involved in that specific rally. The third shot type is also included in this table. The player column will allow us to gain demographic information for every player on the court during a rally. The players handedness and gender will be pulled in as features to train this model. Lastly the shot table will be called to bring in the location of the return, where the this shot will be hit from and how fast the return approached. This shot level analysis can help inform player of which shot to select to give their team the best opportunity to win the point.

Digging into some of the data it is important to understand some of the context of the dataset. First in figure 1 below we can see the number of rally’s won by the serving team. In pickleball scoring teams can only score on serve. And as we see in figure 1 below winning rallies on serve is difficult as the serving team only wins just over 40 percent of rallies. Thus, further emphasizes the importance of proper shot selection as the serving team to win points.

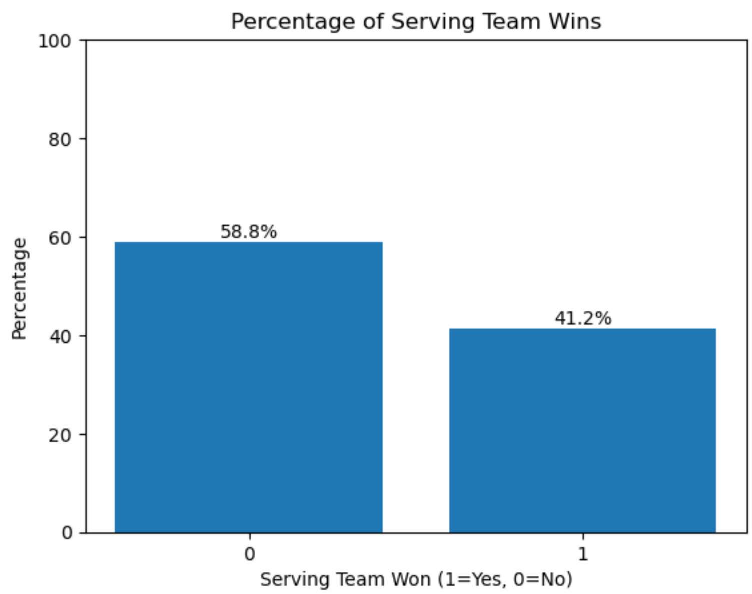

Figure 1: Rally results for serving team.

Next, we evaluated the distribution of third shot types in won points by the serving team. As seen in figure 2 the distribution is relatively even between drive and drop.

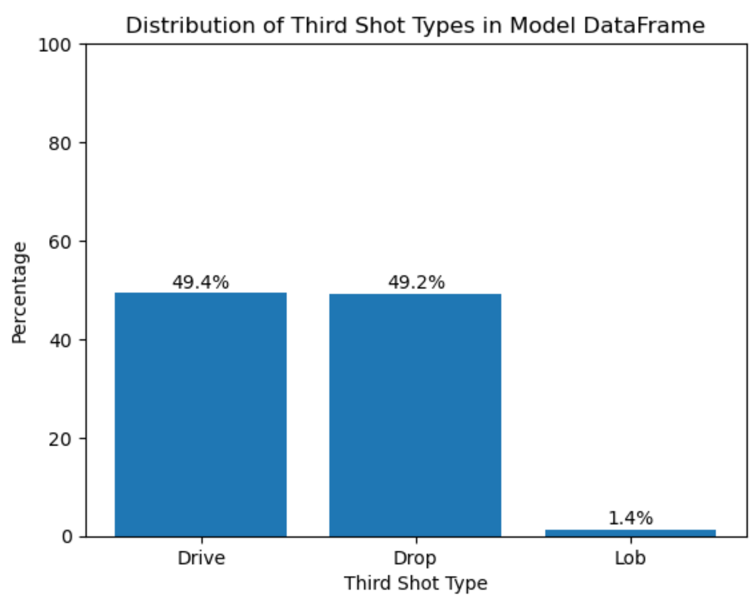

Figure 2: bar graph showing distribution of third shot types in won points.

Methods.

To build a model that well help players develop a shot selection strategy a logistic regression, random forest, XGboost and MLP Classifier (neural network) models were all trained using grid search to select the best parameters and build the most accurate model at selecting the third shot type that led to the serving team winning the rally. These four models were chosen for their ability to evaluate both linear and nonlinear trends. In addition, the machine learning models proved to be very successful in evaluating the complex interactions between player attributes as well as ball speed and location. All models were trained on an 80/20 train test split and evaluated on accuracy of the test set to minimize the effects of overfitting. Of this first and more robust model the most accurate model was the logistic regression classifier with an accuracy of 67.7%. The confusion can be seen below in figure 3.

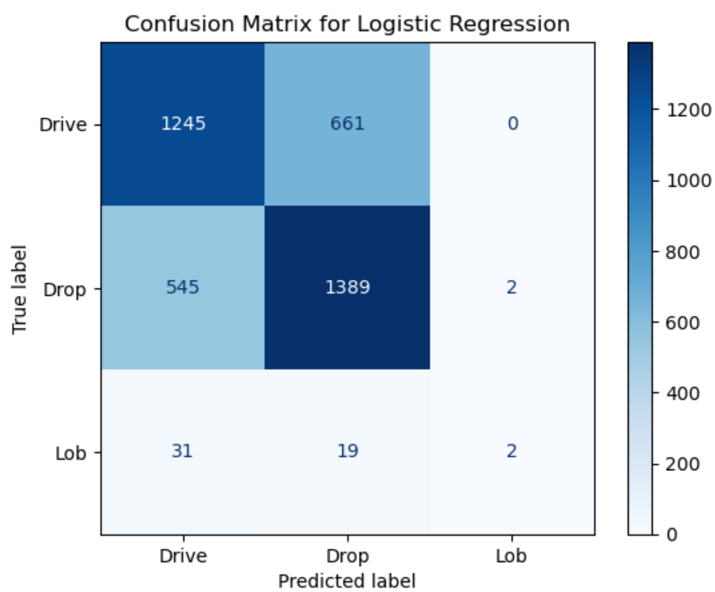

Figure 3: Confusion Matrix: Robust Logistic Regression.

The first model is very complex due to the nature of it being trained with specific opponent data. Having specific opponent data is very unlikely at the amateur ranks as it is unlikely that there are any previous performances for those specific players to train the model off to implement in game. Additionally, that model requires location, ball speed and opponent and team positioning data which can be very difficult to process in real-time for anybody outside the professional ranks. Therefore, A simpler model was trained for amateur implementation using only features easily identifiable player handedness and at what point in the game the players are in. This model was also trained on 80-20 train test split and evaluated on accuracy. The same four models were trained with the best being the XGboost with and accuracy of 54.1% the confusion matrix and feature importance can be seen in figures 4&5 respectively.

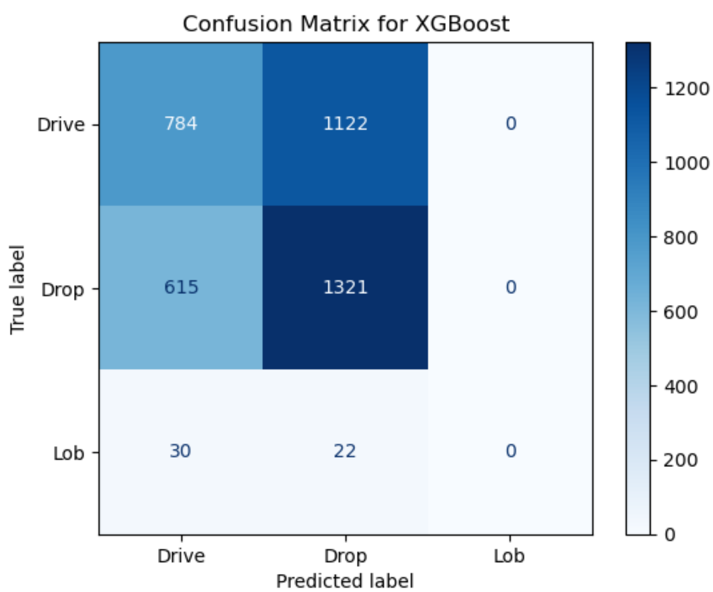

Figure 4: Confusion Matrix: Simplified XGBoost (Amateur Model)

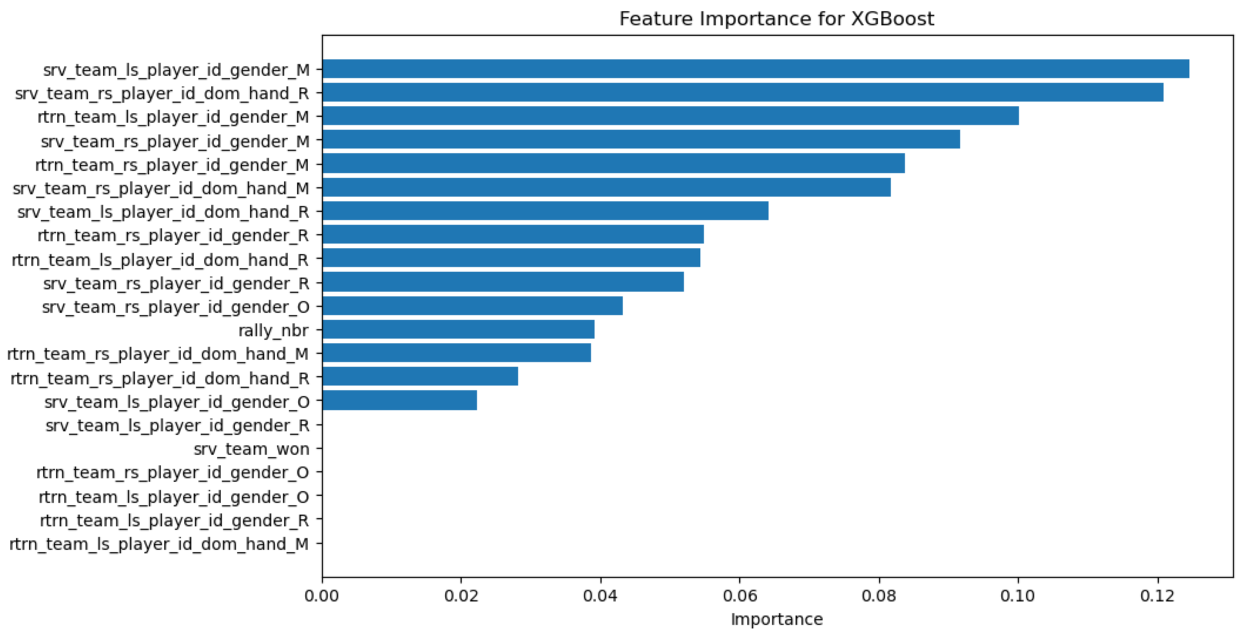

Figure 5: Feature Importance Bar Chart Simplified XGBoost (Amateur Model)

Analysis

Overall, both models perform quite well when being able to accurately predict which shot selection will lead to the serving team win the point. Both the complex model and the simpler model well outperformed the random chance of 33% accuracy in predicting the proper shot selection. When evaluating which features were the most important in predicting success all of the most important features were the handedness and gender of the server and the handedness and gender of the left side opponent. This is expected because in pickleball typically the player on the left side of the court covers most of the court. This is information that a team can simply throw into the model and quickly develop a third shot strategy before a match.

Conclusion

Both models, the more complex and the less complex model developed for amateurs, appear to be relatively effective at helping players select the correct third shot type based on their opponent, opponent demographics and the point they are at in the game. I think there is value in both models being made available to players to help players maximize their performance and get more wins.

Assumptions

Features with missing values in the dataset were replaced with the median response. Having complete data is essential to training these models but there is potential that performance is misrepresented because of these assumptions. That means that when implemented players could lose points due to misinformation.

Limitations

The limitations of this model come from that they are trained using a large data set of diverse players. However, they are all taken from players who care enough about their pickleball performance to pay for the platform for game analysis. Therefore, the promising results that our model see's in this project may not apply to lower levels of play as it is likely that these game types were not included in the dataset that trained the model. This making the model and strategy suggested less effective.

Challenges

Challenges of implementing and using this model are that the model is most effective when it has extensive data involving all the player on the court. Without that we see the model performance suffer. Therefore, using the model to curate a successful strategy at the amateur ranks where in many cases opponents are unknown beforehand and getting game film to train the model is nearly impossible this limits the effectiveness of the model.

Future Uses/ Additional Applications

Due to this model proving to be somewhat effective in selecting shot type for the critical third shot. In the future models could be trained to help layers select strategies for all shots in a match. One that would be interesting and effective is a model predicting where a team should serve (location) given their opponent demographics and positioning to give themselves the best opportunity to win the point. I believe that there is application for these types of models outside the third shot.

Implementation Plan

I think this model should be implemented as a starting point and to inform strategy at both the amateur and professional ranks. The models appeared able to predict winning shot selections with a level of accuracy greater than random. Therefore, I think players should start matches by implementing this strategy and then adjust in game if needed.

Ethical Assessment

There are two main ethical considerations when it comes to evaluating this project. The first being that implementing this advanced level of analytics into recreational sport may cause a change in the perceived enjoyment players as they feel that implementing analytics is essential to compete. The second being privacy as this model rewards players with the data to train and implement, making a huge incentive for players to record opponents matches without consent.

Ten Questions an Audience Would Ask.

1) Does this model consider the location of where the third shot was hit to?
    a. Yes, the professional or more complex model is trained including the location of the return or where the third shot is hit from. This is only included in the professional model because it may be unrealistic for amateurs to implement this in strategy as computing location of the shot adds more complexity the player has to do in point. However, It does not train on where to hit the third shot to. This would be an interesting iteration of this project to add in the future.
2) Does this model consider how good the player is at all 3 shot types?
    a. To some extent yes. This model is trained on which player is performing the third shot but there is not metrics for how skilled the player is at each shot, just how that shot has performed in previous matches.
3) Does this model include opponent's athleticism?
    a. Similar to the question about shot type skill. The model is trained on which player is taking the shot and player athleticism is a component in the outcome of the point. However, there are not variables trained on that measure a player's athletic ability.
4) Does this model account for wind?
    a. No, this model does not account for wind. Hitting specific shot types into or with the wind are considerably easier or more difficult. Assing a wind variable would likely add to the accuracy of the model.
5) Does this model account for ball type?
    a. No, this model does not account for the ball type. Ball type would also likely add to the accuracy of this model. Faster playing balls are considerably more difficult

to “drop”. Because of these discrepancies adding in the ball type is likely to improve the model’s effectiveness for the amateur and professional models.

6) Does this model account for the playing surface?
    a. No, this model does not account for playing surface. This would also be a good addition to the model, however as pickleball continues to be further invested into, the playing surface is likely to become more unanimously tennis hardcourt. In the meantime, there are several other surfaces used such as gym floor as well as roll out surfaces. All these effect the ball bounce and game dynamics. Adding playing surface to training the model will likely improve effectiveness.
7) I have heard about this hybrid shot. Is that shot represented in this project?
    a. No, the Hybrid shot is not denoted separately from the drive in this model. That is a limitation of the data source not recognizing the hybrid shot differently from the drive.
8) Is paddle type or generation accounted for in this model?
    a. No paddle types are not included in the model. Different paddles have different performance capabilities. Different shot types may become easier or more difficult based on the paddle type. For example, powerful paddles are far mor difficult to hit finesse shots like the drop with but produce faster drives. Including this in the model would likely improve performance.
9) Do I need to log my own games to put this model into use?
    a. For the model to be most effective you need to have records of your own game to predict upon. You can predict with only using opponent data however the model performance will likely suffer without previous game data.

10) Do I have to be a Pklmart customer to implement this model into my own games.

    a. No, this model is not proprietary to Pklmart. However, recording enough of your own shot level data to predict on is a very difficult task. In theory however the model is not reliant on the Pklmart database for make game predictions. If one collected enough data on themselves and opponents they could predict on that data and develop a strategy.

Appendix A: Data Dictionary

Rally Table:

Column	Type	Comments
rally_id	varchar	Rally ID
match_id	varchar	Match ID
game_id	varchar	Game ID
point_nbr	int4	Rally number within a game (should really be called rally_nbr)
srv_team_id	varchar	Team ID of the serving team
srv_player_id	varchar	Player ID of the serving player
rtrn_team_id	varchar	Team ID of the returning team
rtrn_player_id	varchar	Player ID of the returning player
ts_player_id	varchar	Player who took the third shot (if a third shot was hit)
ts_type	varchar	Third shot type
w_team_id	varchar	Team ID of the team who won the rally
to_ind	varchar	Timeout indicator
to_team_id	varchar	Team ID of who took the timeout (if a timeout was taken)
rally_len	int4	Number of shots in the rally
srv_switch_ind	varchar	Indicator of if the serving team stacked
rtrn_switch_ind	varchar	Indicator of if the returning team stacked
srv_team_flipped_ind	varchar	Indicator of if the serving team is not in the position they starte
rtrn_team_flipped_ind	varchar	Indicator of if the returning team is not in the position they star stacking)
ending_type	varchar	Indicates if a winner, error, or unforced error ended the rally
ending_player_id	varchar	Player associated with the winner, error, or unforced error that
lob_cnt	int4	Lobs hit in the rally
dink_cnt	int4	Dinks hit in the rally
maint_dtm	timestamptz	
maint_app	varchar	
create_dtm	timestamptz	
create_app	varchar	

Player Table

Column	Type	Comments
player_id	varchar	Unique player ID
first_nm	varchar	Player’s first name
last_nm	varchar	Player’s last name
supp_nm	varchar	Player’s middle or supplemental name
gender	varchar	Gender
dom_hand	varchar	Dominant hand (left or right)
maint_dtm	timestamptz	

maint_app	varchar
create_dtm	timestamptz
create_app	varchar

Shot Table:

Column	Type	Comments
shot_id	varchar	Shot ID
rally_id	varchar	Rally ID
shot_nbr	int4	Shot number (within a rally)
shot_type_orig	varchar	Shot type as entered in the Data Entry Tool
shot_type	varchar	Shot type after applying predictive shot type model
entry_ts	timestamp	Timestamp of when the click to record the shot occurred
btt_before	numeric	Ball travel time between the previous shot and the shot in question (value is
btt_after	numeric	Ball travel time between the shot in question and the subsequent shot (valu
loc_x	numeric	X coordinate, in feet, of where contact was made. A value of 0 implies the si player.
loc_y	numeric	Y coordinate, in feet, of where contact was made. A value of 0 implies conta baseline.
next_loc_x	numeric	X coordinate of the subsequent shot.
next_loc_y	numeric	Y coordinate of the subsequent shot.
model_id	varchar	Model ID that was used to classify the shot_type field, if any
maint_dtm	timestamptz	
maint_app	varchar	
create_dtm	timestamptz	
create_app	varchar	

Appendix B: References

Pickleball + data = 🤯. pklmart. (n.d.). https://pklmart.com/

Pickleball market. Market.us. (2025, May 5). https://market.us/report/pickleball-market/